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RESEARCH

# Genetic Variability of Transpiration Response of Soybean [*Glycine max* (L.) Merr.] Shoots to Leaf Hydraulic Conductance Inhibitor $AgNO_3$

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**ABSTRACT**

Transpiration rate (TR) of a slow-wilting soybean [*Glycine max* (L.) Merr.] line, PI 416937, is constant with increasing vapor pressure deficit (VPD) at high VPD (> ~2 kPa). The basis of such a limitation on TR was recently linked to limited leaf hydraulic conductance. It was hypothesized that this genotype may lack an $AgNO_3$–sensitive protein-mediated water pathway in the leaf, possibly involving aquaporins (AQPs), causing it to have a low hydraulic conductance. The possibility of genetic variability for sensitivity of derooted plants exposed to 200 μM $AgNO_3$ was investigated among 12 soybean genotypes including progeny of PI 416937. Segregation among genotypes in their response to $AgNO_3$ was observed. Genotype PI 416937 and two of its progeny lines were insensitive to the $AgNO_3$ treatment indicating that they lacked a protein-mediated water pathway that is sensitive to this inhibitor. The remaining nine genotypes were sensitive to the $AgNO_3$ treatment, and the decrease in TR indicated that the $AgNO_3$ sensitivity accounted for 25 to 50% of the usual hydraulic pathway in the leaves. Among the $AgNO_3$–sensitive lines, the genotypes could be further segregated based on previous observations in the response of intact plants to increasing VPD. It is hypothesized that this additional segregation might be a result of a difference in hydraulic limitation in their roots.

**Abbreviations:** AQP, aquaporin; DTR, drop in TR; NTR, normalized TR; TR, transpiration rate; $TR_p$, the average of the last four to six TR measurements following the inhibitor treatment; $TR_0$, average of the four TR values before inhibitor treatments; V50, the time at which the TR is halfway between $TR_0$ and $TR_p$; VPD, vapor pressure deficit.

Transpiration is a vital function for plant growth and development. Water flux as a result of transpiration needs to be supplied by hydraulic flow in the plant from the root surfaces to the sites of evaporation in the leaves. The resistance to hydraulic flow in the plant is dispersed in the plant between roots and leaves. The proportion of the total hydraulic resistance in the plant attributed to leaves varies from about 30% to as high as 80 to 98% in some species (Sack et al., 2003; Sack and Holbrook, 2006). The exact nature of such limitations and the link between the leaf hydraulic system and plant transpiration rate (TR) are not fully understood, although it is the subject of considerable current research (Cochard et al., 2004, 2007; Sack et al., 2003, 2004; Tyree et al., 2005; Ye et al., 2008; Hachez et al., 2008; Kim and Steudle, 2009).

Hydraulic resistance in leaves appears to result from water flow both through the apoplast and symplast. The transcellular pathway, defined as a protein-mediated pathway, appears to be particularly important based on biophysical studies (Sack et al., 2004; Ye et al., 2008) and on chemical studies (aquaporin [AQP] gating by inhibitors; Nardini et al., 2005; Cochard et al., 2007). The involvement of protein-mediated pathways allows leaf

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hydraulic resistance to be sensitive to environmental variables such as temperature and light (e.g., Sack et al., 2004; Cochard et al., 2007; Kim and Steudle, 2009).

Recent studies with soybean [*Glycine max* (L.) Merr.] indicated differences exist among genotypes within a single species in their leaf hydraulic resistance. Fletcher et al. (2007) found that soybean genotype PI 416937, which exhibited a slow-wilting phenotype in the field under drought conditions, could not sustain continually increasing TR at high vapor pressure deficit (VPD). In contrast to two other genotypes, TR in PI 416937 became constant at VPD greater than 2 kPa. The results of Sinclair et al. (2008) showed that the inability of PI 416937 to sustain TR at high VPD was associated with a lower hydraulic conductance (i.e., greater resistance) in the water flow pathway from the leaf xylem into the guard cells.

A recent study of Sadok and Sinclair (2010) indicated that the low hydraulic conductance in PI 416937 was associated with the absence of a silver-sensitive protein-mediated water pathway that existed in other genotypes. Although rarely used on plants, a previous study indicated that silver was a potent inhibitor of AQP-based water flow, which may be insensitive to more common inhibitors such as mercurials (Niemietz and Tyerman, 2002). In the study of Sadok and Sinclair (2010), derooted plants were fed $AgNO_3$ as a potential inhibitor of such a hydraulic pathway in the leaves. The TR of derooted PI 416937 plants was insensitive to being fed $AgNO_3$ at concentrations from 10 to 500 μM. In contrast, derooted plants of the commercial cultivar (N01-11136), which when untreated had continually increasing TR over the entire VPD range up to 3.5 kPa, exhibited dramatic decreases in TR 40 min after being fed $AgNO_3$. The TR of N01-11136 was decreased by 43% of its original value as a result of the $AgNO_3$ treatment. Therefore, this result indicated that in N01-11136 the $AgNO_3$–sensitive water pathway accounted for 43% of the hydraulic pathway in the leaves. Comparing the response to $AgNO_3$ for the two genotypes indicated the possibility of substantial differences in the involvement of protein populations—possibly AQPs—associated with the TR response to high VPD.

Since the previous study only compared two genotypes, the extent of differences in the $AgNO_3$–sensitive response among soybean genotypes is unknown. Additional soybean genotypes have been identified in which TR response to high VPD is similar to either N01-11136 (i.e., increasing linearly) or to PI 416937 (i.e., breakpoint; Sadok and Sinclair, 2009a, b). The objective of this study was to characterize the response of these genotypes to an $AgNO_3$ treatment to investigate possible additional genetic sources of differences associated with $AgNO_3$–sensitive water pathway in the leaf. Also, this study was undertaken to determine if the expression of a breakpoint in TR response to high VPD was consistent with the response to $AgNO_3$ treatment.

# MATERIALS AND METHODS

## Genetic Material

Genotypes were selected for this study based on their TR responses to VPD, which were established in previous studies (Fletcher et al., 2007; Sinclair et al., 2008; Sadok and Sinclair, 2009a, b); (Table 1). The genotypes consisted of two slow-wilting plant introductions (PI 416937 and PI 471938; Sloane et al., 1990; Hudak and Patterson, 1995; Pantalone et al., 1999; Carter et al., 2003; King et al., 2009), five commercial lines (Benning, A5959, Biloxi, N73-1102, and N01-11136), and five recombinant inbred lines resulting from a cross of Benning × PI 416937 (BP-59, -78, -160, -211, and -238). Four lines had a linear TR response to VPD with varying slopes over the whole range of tested VPD (PI 471938, N01-11136, BP-211, and BP-238). Eight had a breakpoint in the TR versus VPD relationship: four had a breakpoint occurring at ≈1.4 kPa (A5959, BP-59, -78, and -160) and four had their breakpoint at ≈2 kPa (PI 416937, Benning, Biloxi, and N73-1102; Table 1). Aside from PI 416937, among the other 11 genotypes six were descendants of PI 416937 (Table 1).

## Growth Conditions and Plant Preparation

Seeds were sown in pots filled with 1.5 to 3 kg of compost garden soil (Scotts Miracle-Gro Co., Marysville, OH) containing slow-release fertilizer (1.5 g N $kg^{-1}$, 0.2 g P $kg^{-1}$, 0.8 g K $kg^{-1}$). Three seeds inoculated with *Bradyrhyzobium japonicum* (Nitragin, Inc., Brookfield, WI) were sown in each pot. The plants were grown in a greenhouse at Gainesville, FL (29°38′ N, 82°22′ W), with temperature regulated for a min. temperature of 20°C and max. temperature of 33°C. Pots were watered every 1 to 2 d. Seven to 15 d after sowing, each pot was thinned to one plant. Plants were grown for approximately 3 wk (Table 2) to vegetative stages ranging from V2 to V3, according to the system of Fehr and Caviness (1977). At that time, pots were over-irrigated daily for 2 to 3 d.

On the afternoon of the day before the experiment, two to three replicate plants per genotype were gently removed from the soil and derooted. Derooting was done by cutting the base of the plant stem underwater. Immediately after the cutting, the cut stems of the shoots were placed in 125-mL beakers containing deionized water and placed in a dark room overnight [approximately 14 h at 22.3°C (±0.26 SE)]. The following morning, the shoots were moved from the dark room and transferred to a new set of 125-mL beakers containing fresh deionized water. Laboratory film (Parafilm "M", Pechiney Plastic Packaging, Chicago, IL) was used to seal the stems in the beakers to avoid direct water evaporation. A small hole was made in the film to avoid the development of negative pressure inside the sealed beaker due to water loss.

## $AgNO_3$ Treatment

An $AgNO_3$ treatment was imposed on the shoots inside a 0.10 $m^3$ custom-made plywood, continuously vented test chamber (depth = 39.4 cm, width = 65.8 cm, length = 40.4 cm) with inner walls covered with aluminum foil. The temperature and humidity surrounding the shoots was constant throughout the treatment. Depending on the experiment (Table 2), shoots of two, three, or six plants were placed together inside the test



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**Table 1. Genetic material used in the study.**

| Genotype | Type† | PI 416937 In pedigree‡ | Transpiration rate (TR) response to vapor pressure deficit (VPD)‡ | | | References§ |
|---|---|---|---|---|---|---|
| | | | Form | Slopes | Breakpoint | |
| | | | | mg $H_2O$ $m^{-2}$ $s^{-1}$ $kPa^{-1}$ | kPa | |
| PI 416937 | PI | – | Bilinear | S1 = 30.8, S2 = –5.6 | 2.13 | a |
| PI 471938 | PI | No | Linear | S = 21.0 | – | a |
| Benning | CL | No | Bilinear | S1 = 18.9, S2 = 4.7 | 1.96 | c |
| A5959 | CL | No | Bilinear | S1 = 33.2, S2 = 5.7 | 1.50 | b |
| Biloxi | CL | No | Bilinear | S1 = 23.0, S2 = 9.3 | 1.80 | b |
| N73-1102 | CL | No | Bilinear | S1 = 22.9, S2 = 2.7 | 1.94 | c |
| N01-11136 | CL | Yes | Linear | S = 29.7 | – | c |
| BP-59 | RIL | Yes | Bilinear | S1 = 23.1, S2 = 8.3 | 1.47 | d |
| BP-78 | RIL | Yes | Bilinear | S1 = 30.6, S2 = 9.1 | 1.31 | d |
| BP-160 | RIL | Yes | Bilinear | S1 = 25.7, S2 = 13.6 | 1.23 | d |
| BP-211 | RIL | Yes | Linear | S = 26.1 | – | d |
| BP-238 | RIL | Yes | Linear | S = 13.6 | – | d |

†Line type: PI, plant introduction; CL, commercial line; RIL, recombinant inbred line.

‡Characteristics of TR response to VPD of the selected lines: Slopes of the TR regressions on VPD, where S1 and S2 are the first and second slopes of a bilinear formalism and S is the slope of a linear formalism.

§References: a, Fletcher et al. (2007); b, Sinclair et al. (2008); c, Sadok and Sinclair (2009a); d, Sadok and Sinclair (2009b).

**Table 2. Summary of the experiments.**

| Exp.† | Genotypes | Replicate per genotype | Sowing date | Measurement date‡ | Chamber conditions | |
|---|---|---|---|---|---|---|
| | | | | | VPD§ | Temperature |
| | | | | | kPa | °C |
| Exp. 1 | PI 416937, N01-11136 | 3 | 12 Jan. 2009 | 23 Feb. 2009 | 3.8 ± 0.05‡ | 34.1 ± 0.12 |
| Exp. 2 | PI 471938 | 2 | 5 Mar. 2009 | 31 Mar. 2009 | 3.8 ± 0.02 | 34.9 ± 0.06 |
| Exp. 3 | Benning, BP-211 | 3 | 26 Mar. 2009 | 14 Apr. 2009 | 3.6 ± 0.01 | 34.8 ± 0.04 |
| Exp. 4 | PI 471938 | 3 | 3 Apr. 2009 | 28 Apr. 2009 | 3.7 ± 0.02 | 34.7 ± 0.06 |
| Exp. 5 | N73-1102, BP-59 | 3 | 13 Apr. 2009 | 5 May 2009 | 3.5 ± 0.02 | 34.7 ± 0.05 |
| Exp. 6 | A5959, Biloxi | 3 | 24 Apr. 2009 | 15 May 2009 | 3.6 ± 0.03 | 34.8 ± 0.07 |
| Exp. 7 | BP-78, BP238 | 3 | 8 May 2009 | 29 May 2009 | 3.5 ± 0.02 | 34.8 ± 0.03 |
| Exp. 8 | BP-160 | 3 | 18 May 2009 | 10 June 2009 | 3.6 ± 0.02 | 34.7 ± 0.06 |

†Experiments are presented in chronological order.

‡Mean values ± SE.

§VPD, vapor pressure deficit.

chamber. A stable atmospheric VPD of approximately 3.6 kPa was maintained in the chamber to maximize possible changes in absolute TR in response to the $AgNO_3$ treatment. The stable VPD was achieved by continuously flowing about 40 L $min^{-1}$ of air into the chamber, which resulted in complete air turnover in about 2.5 min. Before entering the chamber, the air was dried by pumping through two polyvinyl chloride (PVC) cylindrical cartridges (diameter = 0.03 m, length = 0.68 m) filled with a silica gel desiccant ($SiO_2$, 10-18 Mesh, S161-500, Fisher Scientific, Fair Lawn, NJ). The air in the chamber was vigorously mixed with two 80-mm diam. fans (ASAF-B83, Cooler Master, Fremont, CA, USA) and three 40-mm diam. fans (EC4010M12C, Evercool Thermal Corp., LTD., San-Chung City, Taiwan ROC). The top of the test chamber was covered with a 5-mm thick Plexiglas sheet above which water-cooled lamps provided a photosynthetic photon flux density at ≈1060 μmol $m^{-2}$ $s^{-1}$ at canopy level. Temperature and relative humidity were measured every 10 min inside the chamber by two pocket humidity–temperature pens (Extech Instruments, League City, TX). Temperature in the test chamber was maintained at 34 to 35°C (temperature amplitude: up to 1.1°C). For soybean, it has been reported that significant effects of temperature on photosynthetic rates or heat shock proteins accumulation only occurred above this temperature range, typically in a range of 36 to 40°C (Key et al., 1985; Campbell et al., 1990; Hsieh et al., 1992; Vu et al., 1997).

Experiments were initiated by acclimatizing the shoots of derooted plants for approximately 80 min in the test chamber until water loss rates were at steady state. Following acclimation, beakers containing the shoots were weighed every 20 min on a balance with a resolution of 0.01 g (Model XP-300, Denver Instrument, Denver, CO). After four consecutive weighings, the shoots were quickly transferred (i.e., in 120–180 s) in darkness and stems sealed into dark brown 30-mL glass bottles (Fisher Scientific, Suwanee, GA) containing 200 μM $AgNO_3$ solutions at room temperature. The 200 μM $AgNO_3$ concentration was selected because this concentration was the lowest concentration resulting in maximum TR decrease in the study with N01-11136 (Sadok and Sinclair, 2010). In addition, Sadok and Sinclair (2010) showed that there was no response in TR to 200 μM nitrate exposure. The solution was prepared 1 to 2 d before the experiments and stored in darkness at 4°C. Simultaneous with the 20 min–step weighings, temperature and humidity conditions in the test chamber were recorded. The



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experiment was terminated when TR following the $AgNO_3$ treatment was constant for four to six consecutive measurements. Measurements of TR for each shoot were generally completed in about 240 min after $AgNO_3$ treatment. Following measurements of TR response to the $AgNO_3$ treatment, leaves were removed from the shoot and leaf area was measured (Leaf Area Meter, model LI-3100, Licor, Lincoln, NE). The leaf area data were used to express TR for each shoot on a leaf area basis.

## Data Analysis

To facilitate comparison between genotypes, a normalized TR (NTR) following the inhibitor treatment was calculated for each genotype as the ratio between the TR value following $AgNO_3$ treatment and the average of the four TR values before inhibitor treatment ($TR_0$).

A Boltzmann sigmoid equation (Eq. [1]) was fitted to the averaged TR across replicates to describe the decrease patterns in TR kinetics following inhibitor treatment of the shoots;

$$TR = TR_P + ([TR_0 - TR_P]/\{1 + \exp[(V50 - Time)/C]\}), \text{ Eq. [1]}$$

where $TR_P$ is the average of the last four to six TR measurements following the inhibitor treatment (see Fig. 1), V50 is the time at which the TR has decreased halfway between $TR_0$ and $TR_P$, and $C$ is a coefficient that represents the steepness of the TR decrease curve over time. In most cases there was not enough data from individual shoots to fit this formalism, so results from all replicate shoots of a genotype were combined in the regressions. All parameters in Eq. [1] were obtained by this regression procedure.

Since the Botlzmann equation was not applied to single replicate plants and to genotypes for which TR was insensitive to $AgNO_3$, determining individual fraction of the initial total TR associated with exposure to $AgNO_3$ was calculated for each shoot from the overall decrease in TR (DTR, %):

$$DTR = [1 - (TR_P/TR_0)] \times 100 = [(TR_0 - TR_P)/TR_0] \times 100, \text{ Eq. [2]}$$

If a genotype was not responsive to the 200 μM $AgNO_3$ treatment with no plateau, the value assumed for $TR_P$ in this case was calculated for the 60- to 100-min period during which a plateau was observed after $AgNO_3$ treatment for the responsive genotypes. For each genotype, DTR (drop in TR) and NTR values were the average of those three to five individual shoots. All statistical analysis (one-way ANOVA with post hoc Tukey HSD test, *t* tests, and correlation tests) and Boltzmann regressions were performed with GraphPad Prism (GraphPad Software Inc., San Diego, CA). Because we did not randomly select the genotypes, the genotypic effects (12 levels) of the ANOVA model were fixed.

# RESULTS

In addition to the results obtained for the 10 genotypes in this study, the original results for genotypes PI 416937 and N01-11136 (Sadok and Sinclair, 2010) are also presented for comparison. Overall, eight experiments were conducted to obtain the TR response to the 200 μM $AgNO_3$ treatment for all twelve genotypes (Table 2). In all experiments except one (Exp. 2), three replicate shoots per genotype were studied simultaneously. As a check for consistency of the responses, experiment Exp. 2 has been repeated (experiment Exp. 4), revealing no significant difference in $TR_0$, $TR_P$, or DTR values (*t* test, $p > 0.41$). Overall, the number of replicate shoots per genotype ranged from three to five.

Comparison of results from the eight experiments required that the chamber environment was similar across experiments. As indicated in Table 2, VPD and temperature values were stable across experiments, averaging 3.6 kPa (±0.04 SE) and 34.7°C (±0.06 SE), respectively. A second condition was a similarity in leaf area among genotypes since the effect of $AgNO_3$ on leaf AQP may depend on leaf area. In all experiments where two genotypes were tested simultaneously, leaf areas were not significantly different, except for experiment Exp. 6 ($p < 0.01$). The resulting correlation between leaf areas of all individuals and their DTR values was weak ($r^2 = 0.14$, Pearson $r = -0.38$, significant at $p < 0.02$) and the correlation resulted from two data at the extremes. Overall, the results indicated little involvement of leaf area among the tested shoots in the differences observed in TR response to $AgNO_3$ treatment among genotypes.

For genotypes sensitive to $AgNO_3$, the decrease in NTR started 20 to 80 min after treatment was initiated. However, TR decrease patterns differed among genotypes as indicated by the variability in V50 values derived from regression from the start point of TR measurement in the chamber, which ranged from 130 min (±16 SE) for BP-211 to 221 min (±6 SE) for N73-1102 (Table 3). For $AgNO_3$–sensitive genotypes, NTR decreased to a plateau after treatment as illustrated in Fig. 1 for genotypes BP-238, Biloxi, and Benning.

The analysis of the NTR results showed that in addition to the previously documented nonresponse of PI 416937, two of its descendants (BP-59 and BP-160) had TR that was insensitive to the 200 μM $AgNO_3$ treatment (Fig. 1 and 2). All three of these genotypes were selected because they exhibited a breakpoint in TR response to VPD. In contrast to insensitivity to the $AgNO_3$ treatment of the three genotypes, the nine remaining genotypes were sensitive to $AgNO_3$ treatment including the second slow-wilting genotype PI 471938 (Fig. 2). There was a significant genotypic effect in DTR (one-way ANOVA, $p < 0.0001$). The DTR values were significantly different across genotypes (Tukey HSD test, $p < 0.05$) ranging from 25% (BP-238) to 50% (Benning; Fig. 2). Among the $AgNO_3$–sensitive genotypes there were four genotypes selected for a breakpoint in their TR versus VPD response and five that did not exhibit a breakpoint (Table 1 and Fig. 3).

There was not a statistical difference in the mean DTR within the $AgNO_3$–sensitive genotypes for those that exhibited a breakpoint in the TR versus VPD response and those that did not (*t* test, $p > 0.43$; Fig. 3). Hence,



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based on both breakpoint differences of intact plants and $AgNO_3$ sensitivity of derooted plants, there were three distinct groups of genotypes: (i) breakpoint and insensitive to $AgNO_3$ treatment, (ii) breakpoint and sensitive to $AgNO_3$ treatment, and (iii) no breakpoint and sensitive to $AgNO_3$ treatment (Fig. 3).

Values of DTR obtained for all individual shoots from all genotypes were found to correlate with $TR_0$ ($r^2 = 0.43$, Pearson $r = 0.66$, significant at $p < 0.0001$; Fig. 4). That is, there was a tendency for those shoots that were able to sustain high TR with the high VPD conditions of this experiment to be more sensitive to $AgNO_3$ treatment. Of course, the fact that all shoots with low DTR had low $TR_0$ was critical in establishing this relationship.

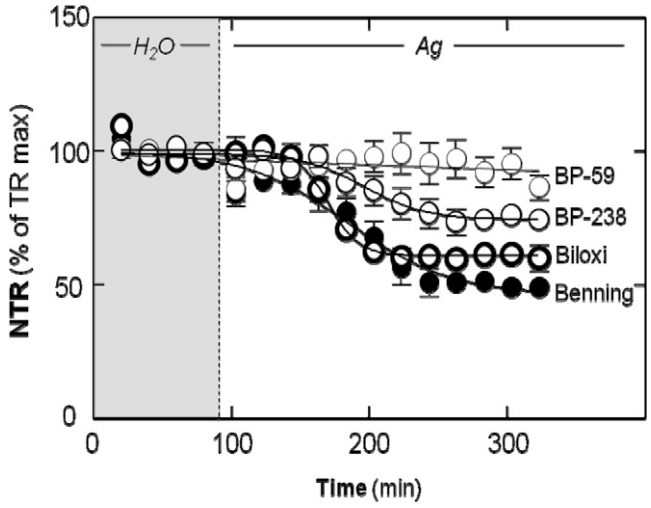



Fig. 1. Examples of time courses of normalized transpiration rate (NTR) response to the 200 μM $AgNO_3$ treatment with constant high vapor pressure deficit (VPD) conditions for two recombinant inbred lines (BP-59 and BP-238) and two commercial lines (Biloxi and Benning). Symbols for the three first lines are open circles with increasing thickness and, for the last line, a closed circle. Gray areas highlight the 80 min sequence where NTR values were measured with deionized water pretreatment. Data were the average of three replicates (± SE). Error bars for some points are invisible when they are smaller than the size of the symbols. Curves are Boltzmann sigmoidal fittings applied to the data of BP-238, Biloxi, and Benning (see Table 3 for derived parameters). Since NTR of BP-59 was insensitive to the $AgNO_3$ treatment, the slope of the NTR versus Time regression was nonsignificant.

# DISCUSSION

## Transpiration Rate Inhibition by $AgNO_3$

Use of AQP inhibitors other than mercurials, which are known to have a large array of side effects, is often advocated (Niemietz and Tyerman, 2002; Maurel et al. (2008); Kaldenhoff et al., 2008). The use of silver, an AQP inhibitor, on plants has been very limited (Volkov et al., 2007; Sadok and Sinclair, 2010). Its exact effects on AQP are not fully understood, but it is believed that, depending on the channel pore size, silver reacts with the sulfhydryl group of a cysteine and also with a histidine, resulting in gating by the targeted AQP (Niemietz and Tyerman, 2002). Silver treatment may also include some side effects such as those due to its interaction with other ions (e.g., $Cl^-$) and other cation binding sites at the cell walls (Niemietz and Tyerman, 2002) or those linked to its interaction with ethylene signaling (Binder et al., 2007). It is not possible to totally discard the side-effect possibility in this study, but the significant correlation between DTR and the control TR values (Fig. 4) indicates much of the response will probably be hydraulic. The simplest hypothesis is that an $AgNO_3$–sensitive AQP population as indicated by a high DTR allows greater water flux with high VPD conditions (estimated by $TR_0$ values). Of course, much of this trend is a consequence of the three genotypes expressing low DTR and low initial TR under the high VPD condition of these experiments.

**Table 3. Transpiration rate (TR) versus time (min) Boltzmann sigmoid fitting results for the genotypes when TR was decreased by 200 μM $AgNO_3$ treatment applied at 80 min.**

| Genotypes† | Boltzmann sigmoid fit‡ | | | | |
|---|---|---|---|---|---|
| | $TR_P$ | $TR_0$ | V50 | C | $R^2$ |
| | % of $TR_0$ | | min | mg $H_2O$ $m^{-2}$ $s^{-1}$ | |
| BP-238 | 74.3 ± 1.5§ | 99.1 ± 1.1 | 196.6 ± 7.0 | –22.4 ± 6.2 | 0.96 |
| BP-78 | 70.8 ± 1.2 | 98.4 ± 1.1 | 183.7 ± 5.1 | –17.9 ± 4.5 | 0.97 |
| PI 471938 | 57.4 ± 11.9 | 105.1 ± 5.9 | 208.9 ± 31.4 | –76.5 ± 36.0 | 0.96 |
| N73-1102 | 67.4 ± 2.0 | 97.9 ± 1.2 | 220.7 ± .6.0 | –17.5 ± 5.2 | 0.96 |
| N01-11136 | 62.2 ± 3.0 | 102.4 ± 3.2 | 169.1 ± 11.1 | –47.5 ± 12.8 | 0.97 |
| A5959 | 62.4 ± 4.2 | 100.9 ± 2.8 | 207.7 ± 11.0 | –18.7 ± 9.6 | 0.88 |
| Biloxi | 60.8 ± 1.4 | 100.4 ± 1.4 | 170.5 ± 3.5 | –11.2 ± 3.0 | 0.98 |
| BP-211 | 49.6 ± 3.7 | 110.6 ± 8.7 | 129.8 ± 16.4 | –50.0 ± 15.6 | 0.96 |
| Benning | 46.3 ± 3.8 | 100.0 ± 3.0 | 184.9 ± 9.3 | –37.5 ± 9.4 | 0.97 |

†Genotypes are presented according to an increasing DTR order (see Fig. 2).

‡The Boltzmann fits were applied to the genotype means of measurements collected during the time course of an experiment, where $TR_P$ = plateau TR after inhibitor treatment, $TR_0$, = TR before inhibitor treatment, V50 = time when TR decreased halfway between $TR_0$ and $TR_P$, and *C* = coefficient that represents the steepness of the TR decrease curve over time.

§Parameter estimate ± SE.



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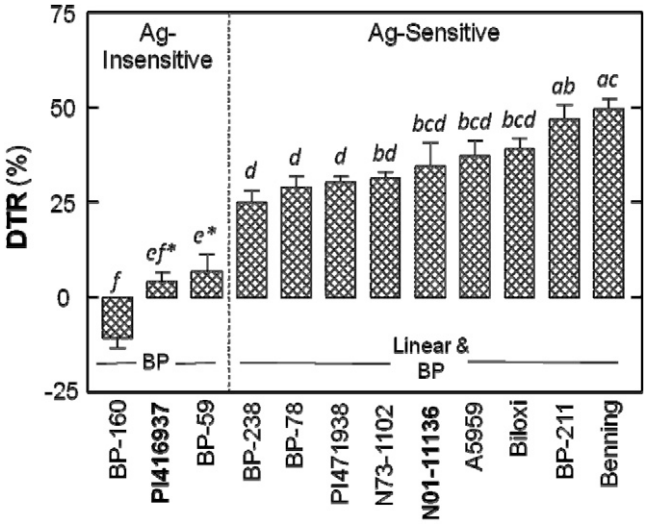



Fig. 2. Genetic variability of silver-induced drop in transpiration rate (DTR) among the 12 studied genotypes. Drop in TR values are plotted in an increasing order. Data are the mean of 3 to 5 replicate plants and error bars are the SE of the mean. Significant differences among genotype means was evaluated by the Tukey multiple comparisons test. Bars with different letters are significantly different ($p < 0.05$). *Denotes nonsignificant difference from a DTR of 0. 'Linear' and 'BP' refer to lines without and with breakpoints, respectively, in their transpiration rate (TR) versus vapor pressure deficit (VPD) relationship. Previously studied genotypes (PI 416937 and N01-11136; Sadok and Sinclair 2010) are highlighted in bold characters.

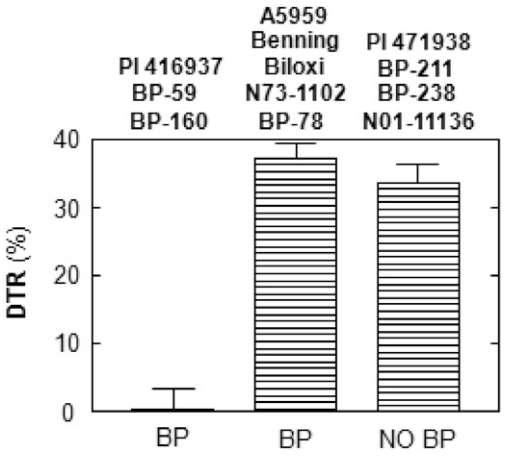



Fig. 3. Variability of silver-induced drop in transpiration rate (DTR) between groups of genotypes with a breakpoint in their transpiration rate (TR) response to vapor pressure deficit (VPD) and the group of lines in which TR responded linearly to VPD (no breakpoint). The lines belonging to each group are indicated above the corresponding bars. Error bars are the SE of the mean.

No matter the mode of action, the results of this study demonstrated substantial variation among soybean genotypes for their response to the $AgNO_3$ treatment (Fig. 2). Nine genotypes were sensitive to the $AgNO_3$ treatment indicating that a segment of their leaf water flow was vulnerable to silver inhibition. The extent of the decrease in NTR following $AgNO_3$ treatment indicated that 25 to 50% of water flow in

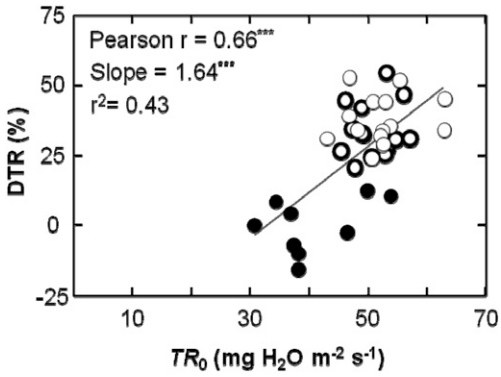



Fig. 4. Relationship between individual control transpiration rate (TR) values ($TR_0$) and their corresponding drop in TR (DTR) measurements across the 38 replicate shoots of the 12 studied genotypes. The three symbols refer to the three groups of lines reported in Fig. 3. Closed circles are the lines with breakpoints in which TR was insensitive to 200 μM $AgNO_3$ treatment. Open thin and thick circles are the lines respectively with and without breakpoint, in which TR was sensitive to 200 μM $AgNO_3$ treatment. Pearson *r*, the slope, and $R^2$ of the relationship are given. ***Denotes $p < 0.0001$.

leaves was associated with silver-inhibited protein-mediated pathways, possibly AQPs. Such possibility is reinforced by the results of MacRobbie (2006), which indicated the existence of putative tonoplast, silver-sensitive AQP in the guard cells of *Commelina communis* L. Although previous studies have established the importance of the trans-cellular pathway in leaf hydraulics for various species (Sack et al., 2004; Nardini et al., 2005; Tyree et al., 2005; Cochard et al., 2007; Ye et al., 2008), these results with soybean genotypes indicate there may be intraspecific variability either in the AQP populations associated with transcellular flow or in the relative importance of such a pathway.

## Implications of the Genetic Variability of Transpiration Rate Response to Silver

This study identified three distinct groups of soybean genotypes characterized by the TR response to VPD and by their response to $AgNO_3$ treatment (Fig. 3). The TR results for one group of three genotypes—PI 416937, BP-59, and BP-160—were insensitive to $AgNO_3$ treatment. These three genotypes were previously identified as exhibiting a breakpoint in TR with increasing VPD (Sadok and Sinclair, 2009b). Further, the breakpoint in PI 416937 was shown to be a result of limited hydraulic conductance in its leaves (Sinclair et al., 2008). All of these results are consistent with a hypothesis that PI 416937 is lacking an $AgNO_3$–sensitive hydraulic pathway in its leaves. Without such a pathway, water flow in leaves is restricted causing the plant to have limited TR at high VPD and, consequently, conserving water in the soil under field conditions to result in the slow-wilting phenotype (King et al., 2009). Such a hypothesis is reinforced by the recent results of Charlson et al. (2009) showing that one quantitative trait locus (QTL) associated



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with canopy wilting in soybean mapped closely to a locus harboring an AQP gene.

A second group of five genotypes originally identified as having a breakpoint in TR in response to increasing VPD were found to be sensitive to $AgNO_3$. Two of these lines (A5959 and Biloxi) were included in the study of Sinclair et al. (2008) and found to have only a modest limitation to hydraulic flow in their leaves. It is hypothesized that this second group of genotypes may have a silver-sensitive water pathway in their leaves, possibly involving a population of $AgNO_3$–sensitive AQP. Therefore, the observation of breakpoint for the intact plants of these genotypes requires an explanation beyond a limitation in restricted leaf hydraulics. One possibility is that this group of genotypes has a limited conductance in the hydraulic pathway below the leaves. A candidate for such a limited conductance could be in the roots. Because there is variation in the VPD at which the breakpoint occurred in intact plants among these genotypes (Table 1), it is possible that root conductance limitation may vary among these genotypes. Certainly, these five genotypes are good candidates for exploring intraspecific variation in root hydraulic conductance.

The third group of four genotypes were also sensitive to the $AgNO_3$–treatment but were selected for study because these intact plants exhibited no breakpoint in their TR response to increasing VPD. The results of the $AgNO_3$–treatment indicate that these genotypes have a $AgNO_3$–sensitive pathway in their leaves allowing higher hydraulic conductance. The fact that they exhibited no breakpoint in intact plants indicates that there is no major conductance limitation at any point in the plant in their hydraulic pathway. Their unrestricted hydraulic pathway either in the leaves or roots allows them to have high TR across the entire range of tested VPD from 1 to 3.5 kPa.

The results for the five genotypes that are progeny of the Benning × PI 416937 mating indicate that inheritance of hydraulic conductance may be complex, mirroring recent findings that revealed that canopy wilting inheritance was polygenic (Charlson et al., 2009). The results from the experiments presented here indicated that for Benning, the hydraulic conductance in the leaves is relatively high but that there may be a root conductance limitation, while PI 416937 has a substantial hydraulic limitation in the leaves. The relatively high VPD for the breakpoint of PI 416937 and the results of Sinclair et al. (2008) indicate that it does not have an additional root conductance limitation. Progeny of these two genotypes were identified in each of the three groups identified in this study (Fig. 3). The breakpoint for intact plants of BP-59 and BP-160 was less than that of PI 416937 (Table 1), indicating that these two progeny lines might have hydraulic limitations both in the leaves and roots. Genotype BP-78 was like its Benning parent in that it was $AgNO_3$–sensitive but the intact plant exhibited a breakpoint. The most curious genotypes were BP-211 and BP-238, which were in the third group of genotypes and apparently had little hydraulic limitation either in the leaves or roots.

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